Assignment 7: Read Sensor Values Using Arduino IDE

Internship: IoT with AI

Week: 1

Topic: Introduction to IoT and Electronics Basics

1. Introduction

Sensors are important components of Internet of Things (IoT) systems. They collect information from the surrounding environment and convert physical parameters into electrical signals.

Arduino can interface with different sensors, read their values and process the collected data. These readings can be displayed using the Serial Monitor for observation and analysis.

In this experiment, a TMP36 temperature sensor is connected to Arduino Uno to measure the surrounding temperature.

2. Objective

To interface a TMP36 temperature sensor with Arduino Uno, read analog sensor values and display the calculated temperature using the Serial Monitor.

3. Components Required

1. Arduino Uno R3 – 1
2. TMP36 temperature sensor – 1
3. Breadboard – 1
4. Jumper wires – 3
5. Tinkercad Circuits – Online simulation platform

4. Circuit Connections

The TMP36 temperature sensor has three terminals.

- Left pin (VCC): Connected to Arduino 5V.
- Middle pin (Output): Connected to Arduino analog pin A0.
- Right pin (GND): Connected to Arduino GND.

The sensor should be positioned with its flat face towards you and its pins pointing downward when identifying the terminals.

5. Working Principle

The TMP36 is an analog temperature sensor that produces an output voltage corresponding to the surrounding temperature.

The working process is as follows:

1. The TMP36 sensor detects the surrounding temperature.
2. It generates an analog voltage based on the measured temperature.
3. Arduino reads the sensor output through analog pin A0.
4. The analogRead() function converts the voltage into a digital reading.
5. Arduino calculates the corresponding temperature in degrees Celsius.
6. The calculated temperature is displayed using the Serial Monitor.

6. Arduino Program

```
const int sensorPin = A0;

void setup() {
  Serial.begin(9600);
}
```

```
void loop() {
  int sensorValue = analogRead(sensorPin);

  float voltage = sensorValue * (5.0 / 1023.0);

  float temperatureC = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" °C");

  delay(1000);
}
```

7. Simulation Procedure

1. Open Tinkercad Circuits.
2. Create a new circuit and add Arduino Uno.
3. Add a TMP36 temperature sensor to the breadboard.
4. Connect the sensor VCC pin to Arduino 5V.
5. Connect the sensor output pin to analog pin A0.
6. Connect the sensor GND pin to Arduino GND.
7. Open the Code editor and select Text mode.
8. Enter the Arduino program.
9. Click Start Simulation.
10. Open the Serial Monitor and observe the temperature readings.

8. Expected Output

The temperature readings are displayed in the Serial Monitor in degrees Celsius.

Example output:

Temperature: 25.00 °C

Temperature: 26.00 °C

Temperature: 27.00 °C

Note: These are illustrative values. Actual readings depend on the simulated temperature and sensor output.

9. Applications

1. Smart home temperature monitoring.
2. Weather monitoring systems.
3. Industrial temperature monitoring.
4. Greenhouse automation.
5. Environmental monitoring systems.

10. Conclusion

This experiment demonstrates how a TMP36 temperature sensor can be interfaced with Arduino Uno to measure environmental temperature. The sensor generates an analog voltage, which is read and processed by Arduino. The calculated temperature is displayed through the Serial Monitor. This experiment provides practical understanding of analog inputs, sensor interfacing and Arduino programming.